



Cole's Software Solutions

Software Solution

MJ Logistics Gaming Company Software Solution

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WESTERN GOVERNORS UNIVERSITY®

CONTENTS

A.	Introduction	3
A1.	Introduction and Purpose Statement.....	3
A2.	Overview of the Problems	3
A3.	Goals and Objectives	3
A4.	Prerequisites.....	4
A5.	Scope	4
A6.	Environment.....	5
B.	Requirements.....	6
	Business Requirements: REPORTING	6
	User Requirements: OS & BROWSER SUPPORT	6
	Functional Requirements: ORDER MANAGEMENT	6
	Non-Functional Requirements: QUOTING	7
C.	Software Development Methodology.....	7
C1.	Advantages and Disadvantages	7
	Advantages of the Agile Method.....	7
	Disadvantages of the Agile Method	7
	Advantages of the Waterfall Method	8
	Disadvantages of the Waterfall Method	8
C2.	Best suited	8
D.	Create Two Representations of the Software Solution.....	9
	Representation 1.....	9
	Representation 2.....	10
E.	Testing.....	13
	Report Creation	13
	System Accessibility.....	13
	Ticket Logging	14
F.	Sources.....	16
	References	16



A. INTRODUCTION

Cole's Software Solutions (CSS) is proposing a well-rounded and effective customer relationship management (CRM) system for MJ Logistics Gaming Company (MJLGC). The details of the solution are outlined in sections A through E of the document.

A1. INTRODUCTION AND PURPOSE STATEMENT

We at CSS plan to provide MJ Gaming Company with a customizable CRM system that would be developed specifically to meet its distinct requirements. The purpose of the proposed system is to modernize MJLGC's infrastructure by connecting disjointed processes, maximizing data management efficiency, and streamlining sales procedures. Due to the nature of MJLGC's growth, the system will prioritize scalability, as well as access to key features for both onsite and remote employees.

A2. OVERVIEW OF THE PROBLEMS

MJ Gaming Company has outgrown its current system and is seeking a CRM that meets its requirements while retaining as much of the original infrastructure as possible. The aforementioned system consists of disconnected, custom-built spreadsheet and database management software. The current design is an unruly mix of manual and automated processes, lacking the efficiency required to keep pace with MJGC's ongoing business success. Tools and features are not consolidated, readily available, or functional for employees spread across multiple locations and using varying machines.

Cole's Software Solutions is proposing a CRM that will address these issues by providing a centralized, accessible system. The solution will integrate all customer, ticket, and communication data into a unifying system. Manual steps will be replaced with automatic processes, reducing redundancy and risk of human error. The system will implement a comprehensive interface that manages all the CRM's functions, minimizing the time and effort spent on maintaining and switching between different programs. The cloud-based design will ensure that users have access to all tools and data from any approved device, regardless of their location. These solutions will provide MJ Logistics Gaming Company with a modern, scalable, and effective system to support its ongoing growth.

A3. GOALS AND OBJECTIVES

Goals:

- Consolidation – Consolidate contacts, sales information, and reports to one centralized, easy-to-read dashboard
- Access – Ensure employees' access to features is consistent regardless of location or device
- Scalability – Establish a system that can match the growth of MJ Logistics and allows for future enhancements

Objectives:

- Dashboard – Develop a dashboard that is intuitive, user-friendly, and contains all necessary information



- Support – Identify support and maintenance processes, as well as provide a roadmap for agreed-upon future updates
- Documentation – Clearly document the development process to be referenced for future updates or modifications
- Sales – Streamline the entire order management process from quoting to tracking to completing a sale
- Data Management – Manage data associated with meetings between clients, stakeholders, and internal parties. Manage data associated with sales, including forecasting and opportunity analysis. Provide a comprehensive reporting system to display data gathered by both avenues

A4. PREREQUISITES

Number	Prerequisite	Description	Completion Date
1	Review & Response	MJ Logistics Gaming Company reviews Cole's Software Solution CRM proposal. Respond with a counteroffer or non-negotiable requirements not met by the original proposal	11/01/2025
2	Confirmation & Signing	Draft a contract that outlines the details and expectations of the project. Sign the contract once both parties agree upon the terms	11/08/2025
3	Training	Train MJGC employees on system use and functions	11/15/2025
4	Testing & Debugging	Test the system's functionality and ensure that all requirements are met. Identify and troubleshoot initial bugs	11/29/2025
5	Data Transfer	Transfer all data from the previous system to the new system	12/06/2025
6	Live Deployment & Debugging	Deploy the new CRM system. Establish a maintenance schedule. Review and debug the live system	12/13/2025

A5. SCOPE

In-Scope:

- Core CRM Features
 - Report Generation System



- Sales Tracking
 - Ticketing System
 - Order Management System
 - Quoting system
 - Contact Management System
- Interactive Dashboard
- Access Controls
- System Documentation

Out-of-Scope:

- Game development
- Pre-Written Report Queries
- Hardware Replacements or Upgrades

A6. ENVIRONMENT**Operating Systems:**

- IOS
- Windows
- Android

Browsers:

- Google Chrome
- Microsoft Edge
- Apple's Safari
- Firefox
- Other Chromium-Based Browsers

Back-End:

- Cloud-Based Infrastructure hosted on Amazon Web Services (AWS)
- MySQL database management system for secure data storage and querying
- Integration with Microsoft Outlook and the company's existing Active Directory for user authentication and communication management

The Service Level Agreement (SLA) outlines the expectations and commitments of Cole's Software Solutions CRM. It will ensure that the system remains stable, secure, and responsive while in operation. The SLA covers key areas, including connectivity outages, upgrades and custom development, support and maintenance, and pre-deployment testing.

Service Level Agreement:

- Connectivity Outages:
 - The system will maintain a minimum uptime of 99.9%, excluding scheduled updates
 - Redundancy protection will minimize downtime through failover systems, data replication, and multiple infrastructures



- Progress updates will be provided during unexpected outages until service is restored
- Upgrades & Custom Development:
 - Since this CRM is a custom-built solution explicitly designed for MJ Logistics Gaming Company, no upgrades are available
 - Any new feature or significant changes will require a new agreement before new custom development begins
- Support & Maintenance:
 - Standard technical support will be available during business hours, while emergency support for more critical issues will be available 24/7
 - Maintenance will be scheduled during off-peak time to minimize disruption
 - Routine system maintenance will be done to keep the CRM stable and secure
- Pre-Deployment Testing
 - Testing will be handled in a separate secure environment before deployment
 - The testing environment will be a mirror of the live system to ensure full functionality
 - Tests will be conducted on all updates, patches, and new features to verify stability before release

B. REQUIREMENTS

Below, we will summarize the requirements outlined by MJ Logistics Gaming Company that our system covers.

- Reporting
- OS and Browser Support
- Order Management
- Quoting

BUSINESS REQUIREMENTS: REPORTING

Our system's CRM is capable of generating customizable and preset reports, which are displayed on a user-friendly dashboard. Users can filter, format, query, and export data directly from the dashboard. Displayed data will be relevant to the end-user in terms of their department or function. Reports and filters can be saved for future use, reducing the time spent on repetitive tasks. These features will allow employees to create detailed and higher-level reports.

USER REQUIREMENTS: OS & BROWSER SUPPORT

The CSS CRM is designed to work across multiple platforms, so employees can access it regardless of where they work. The system supports the latest versions of desktop and mobile operating systems, including iOS, Microsoft Windows, and Android. It is also compatible with up-to-date versions of Safari, Firefox, Microsoft Edge, Chrome, and other Chromium-based web browsers. This support ensures that employees can use the system using their preferred method without facing compatibility issues.

FUNCTIONAL REQUIREMENTS: ORDER MANAGEMENT



Cole's Software Solution system will assist users in the order management process. First, the system will take orders with or without a preliminary quote. These orders will include tracking information, ensuring that both employees and customers remain informed throughout the process. For repeat customers, a reordering function is available to streamline the transaction process. Part ordering will allow employees to manage the stock of merchandise. Clients will have access to a front-end portal where they can place and track orders, view their purchase history or receipts, or request support for any issues.

NON-FUNCTIONAL REQUIREMENTS: QUOTING

The CRM system provides a comprehensive quoting process. It handles monetary features such as discounts, taxes, freight costs, and different currency types. Supply-related features include inventory and product availability, substitutions, and a pricing catalog. Quote finalization features include electronic signatures, shipping forecasts, and quote delivery. Overall, the provided system supports every step of the quoting process, from creation to finalization.

C. SOFTWARE DEVELOPMENT METHODOLOGY

Below, we will discuss the advantages and disadvantages of the Agile and Waterfall Methodologies.

C1. ADVANTAGES AND DISADVANTAGES

ADVANTAGES OF THE AGILE METHOD

- **Continuous Delivery:** Agile methodology promotes working in short iterations. This style of continuous updating would work well with a gaming company that must release regular updates for its games.
- **Continuous Collaboration:** Agile's focus on short interval releases means there are more opportunities for client or sponsor feedback. This allows developers to change the direction of a product early on rather than scrapping a fully developed product.
- **Technical Excellence:** Agile methodology focuses on simplicity, reliability, and testability of its products. This means that when MJ Logistics releases a product, its clients can count on it to be functional.

(Haunts, Agile Fundamentals - Advantages of Agile, 2020)

DISADVANTAGES OF THE AGILE METHOD

- **Scope Creep:** With Agile's short iterations and repetitive development feedback loop, the original goals of the project can get buried by new goals or features over the product's lifecycle. Backlogging the originally proposed requirements can strain the relationship and trust between the developer and client.
- **Initial Estimates:** Because the product can change over time, it is tough to estimate the time and monetary cost the project requires accurately. This can drain resources and funding, potentially dampening the end quality or sinking a project entirely.



- **Intensity on Team:** Agile's consistent and intense release schedule can wear down developers and staff. This can also weaken Agile's collaboration aspect because it becomes more difficult to keep consistent communication as a project develops or the workload increases.
(Haunts, Agile Fundamentals - Disadvantages of Agile, 2020)

ADVANTAGES OF THE WATERFALL METHOD

- **Well-Defined Stages:** The Waterfall method begins its process by clearly defining each step of the project. The steps are completed in an easy-to-follow, sequential manner, with the output from one step serving as the input for the next.
- **Management:** Because the process is well defined and the requirements are outlined in the beginning. It is easy for a project manager to allocate their team where necessary and plan for the entire project.
- **Departmentalization:** The Waterfall method offers the ability to departmentalize its tasks. This means that each step does not rely on the work of other departments and can progress independently once it is their turn to work. This can help speed up the project due to the reduced reliance on external sources.
(Haunts, Agile Fundamentals - Advantages and Disadvantages of Waterfall, 2020)

DISADVANTAGES OF THE WATERFALL METHOD

- **Lack of Revision:** Due to the nature of processing through a project using the Waterfall method, there is little allowance for changing a project's features or adding requirements. This would limit MJ Logistics' ability to introduce creative ideas later in a product's life cycle.
- **Time schedule:** The Waterfall method must wait until the prior step is completed before moving on. This means that if the preceding step is running behind schedule, it can limit the time available for later steps.
- **Late-Stage Functionality:** A functional product will not be delivered until the last steps of the method. This means that consumers are unable to provide feedback until it is too late to make any changes.
(Haunts, Agile Fundamentals - Advantages and Disadvantages of Waterfall, 2020)

C2. BEST SUITED

Cole's Software Solutions believes that Agile is a better-suited methodology for implementing the requirements briefed by MJ Logistics Gaming Company. One reason is that this is a massive undertaking to revamp an outdated system, and there is a high likelihood that changes will occur during the project's course. Another reason is that the Agile method allows the client to test the product throughout the process and provide feedback. This means that the end-product will be more tailored to what the client wants, rather than what they initially thought they needed. The final reason is that MJ Logistics has employees based in several offices as well as remote workers. With this method's focus on collaboration, issues faced by those employees will be addressed as the project progresses. The Waterfall method, although more structured, is better suited for smaller projects and those where the original requirements are not likely to change.



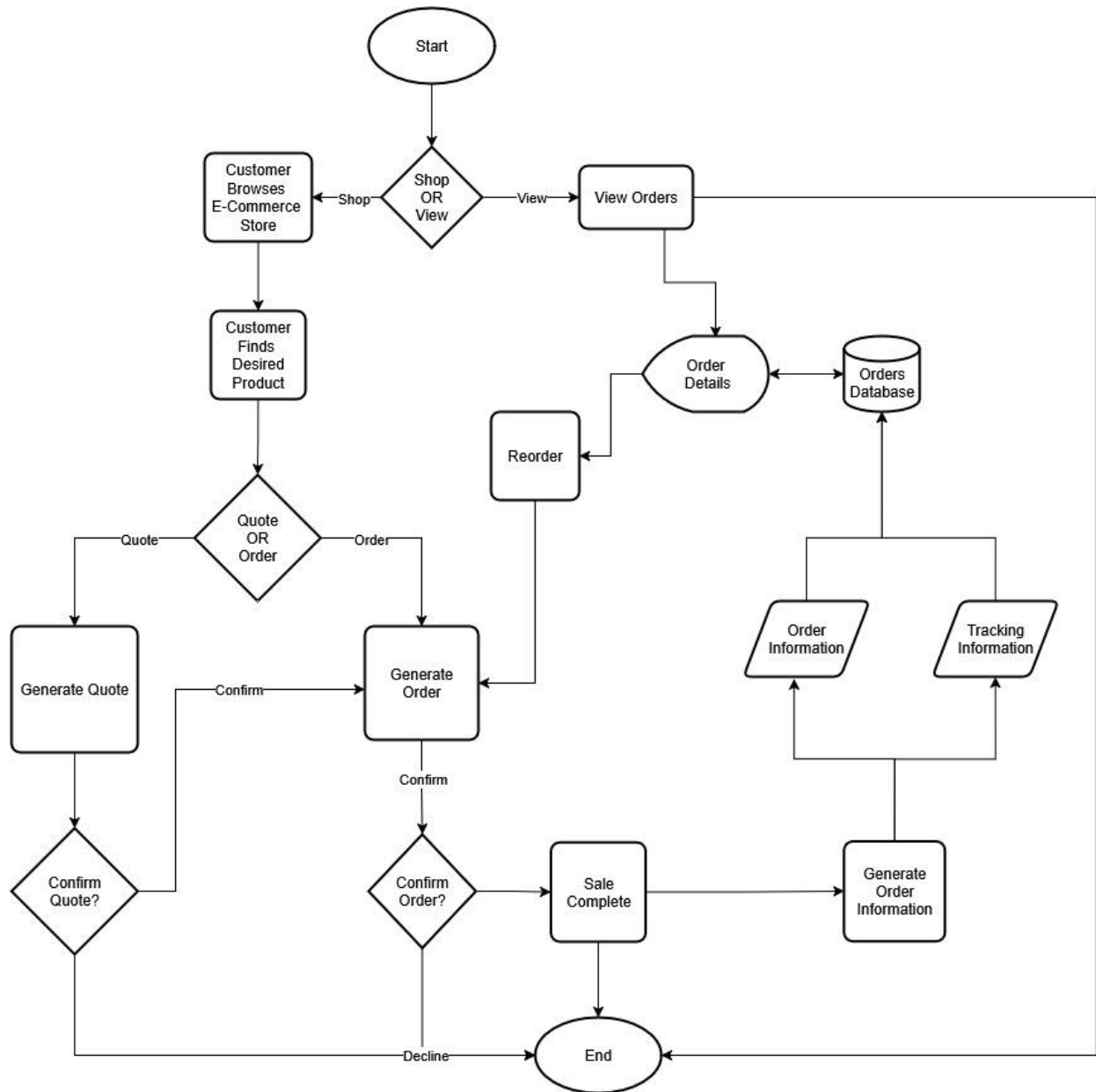
D. CREATE TWO REPRESENTATIONS OF THE SOFTWARE SOLUTION

Below are two visual representations of the proposed CRM software solution. The first is a flow chart illustrating the order management system, and the second is an ERD diagram portraying the ticketing system. Both of these depictions will give a brief insight into how some of the proposed systems will function.

REPRESENTATION 1

Below is a flowchart of the order management system. The System is broken into two major sections, the first being a shopping section where orders and quotes are created. If a quote is accepted, it is turned into an order. Once the order is received, the sale is completed, and the order and tracking information are generated and logged in the orders database. The second section is the order history side, where details and tracking information are displayed. The Order details are taken from the order database. From here, users can reorder previous orders and skip directly to the generate order process instead of browsing the e-commerce store again.





REPRESENTATION 2

Below is an Entity Relationship Diagram (ERD) of the CRM's ticketing system.

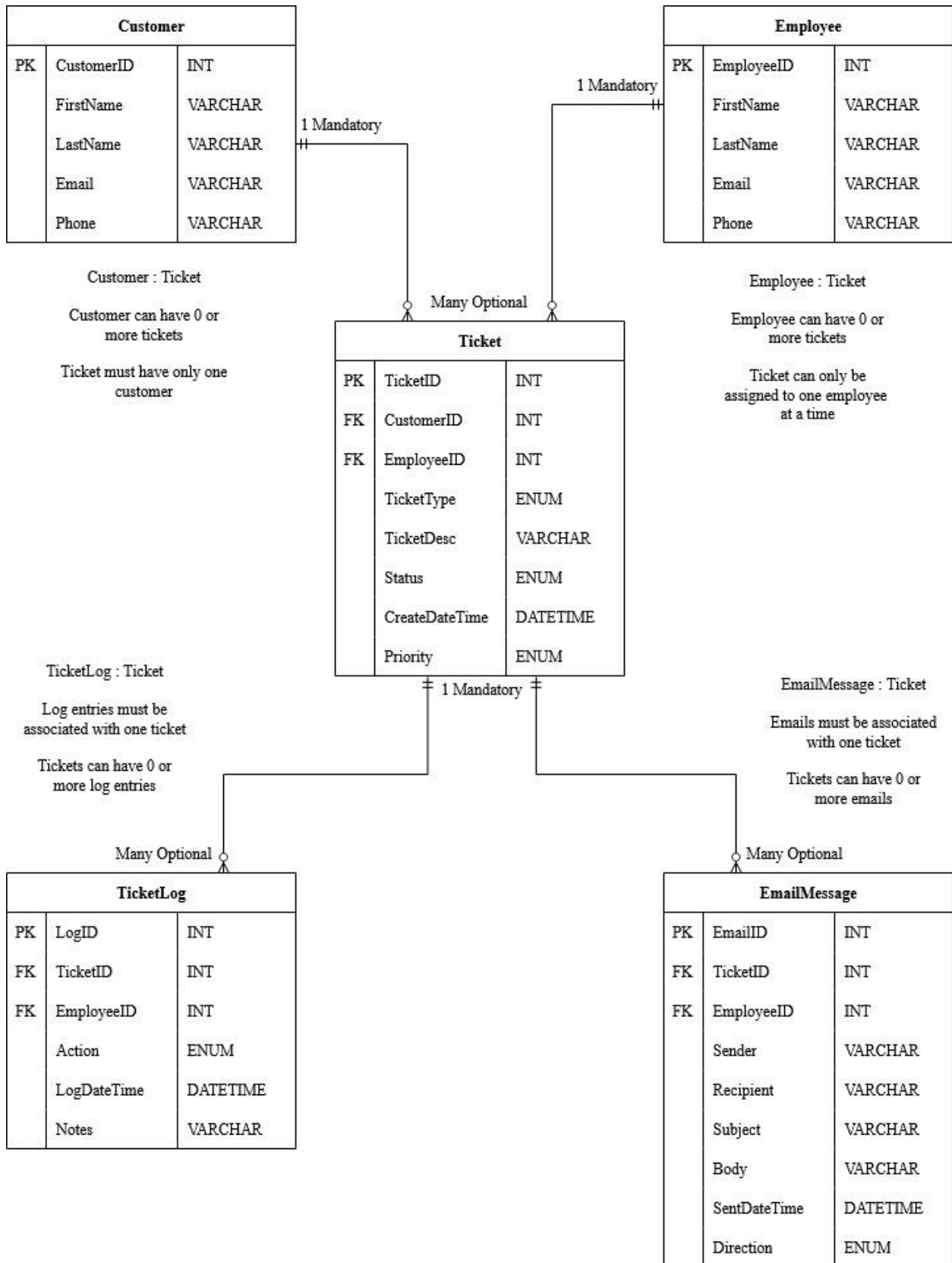
- Customer : Ticket
 - 1 mandatory : many optional



- Each customer can have zero or more tickets linked to their ID
 - A ticket must have only one customer assigned to it
- Employee : Ticket
 - 1 mandatory : many optional
 - Each employee can have zero or more tickets assigned to them
 - A ticket must have exactly one employee working on it at a time
- Ticket : TicketLog
 - 1 mandatory : many optional
 - Each ticket can have zero or more log entries tracking actions
 - Each log must belong to exactly one ticket
- Ticket : EmailMessage
 - 1 mandatory : many optional
 - Each ticket can have zero or more emails associated with it
 - Each email must belong to exactly one ticket

This system would allow for the tracking of communication and information associated with tickets. The ticket entity will record the customer who created it, the type, a brief description, the status, the date and time it was created, and its priority. Customer and Employee entities will keep track of communication information. The TicketLog entity will maintain a workflow and audit trail by logging actions taken, including the date and time, and providing extra notes as necessary. The EmailMessage entity will document follow-up communications by maintaining a transcript of emails, including message details, sender, and recipient information.





E. TESTING**REPORT CREATION****Requirement to be tested:**

The system will generate custom reports and allow for filtering, formatting, querying, saving, and exporting of data.

Preconditions:

- A “Test” dataset that contains valid ticket, employee, and customer data
- The user is logged in with permissions to create and view reports
- The Reports module is accessible

Steps:

1. Log in to the CRM program
2. Navigate to the “Reports” section of the CRM dashboard
3. Select “Create New Report”
4. Choose “Custom Report” and select the “Test” dataset
5. Apply filters: Status = “Closed”, EmployeeID = 1234
6. Add Fields to the report: TicketID, Customer.Email, Customer.FirstName, Customer.LastName, CreateDateTime
7. Sort by CreateDateTime descending
8. Click “Generate Report”
9. Verify data is valid, present, and matches filtered criteria and formatting
10. Click “Save Report” and save as “1234 Closed Ticket Report”
11. Reopen “1234 Closed Ticket Report” and verify that the data is identical
12. Export final report and verify data and formatting consistency

Expected results:

- Report data matches applied filters
- Report displays all required columns in correct order
- Report layout matches formatting
- Report can be saved, retrieved, and exported successfully

Pass/Fail:

Pass: The report is generated successfully using the selected filters. The data is correct and complete. The data appears in the correct order and formatting, matching the chosen columns and sorting. The report can be saved, reopened, and exported without any errors.

SYSTEM ACCESSIBILITY**Requirement to be tested:**

The system must be accessible and fully functional across multiple employee devices and varying known employee locations.



Preconditions:

- A test user with valid employee credentials
- List of approved devices (desktop, laptop, mobile, tablet)
- List of approved operating systems (Windows, macOS, Android, and iOS)
- Secure VPN connection for remote user testing
- Access to all modules of CRM
- A configured virtual machine environment with profiles for all approved devices and operating systems

Steps:

1. Log in to the CRM using a virtual machine configured with profiles for each approved on-site device and operating system:
 - a. Desktop – Windows, macOS
 - b. Laptop – Windows, macOS
 - c. Tablet – iOS, Android
 - d. Mobile – iOS, Android
2. Log in to the CRM using the same virtual machine profiles as “Step 1” using a VPN connection to simulate remote access.
3. Verify that connectivity is successful and the login process completes without errors
4. On each device, both on-site and remote, navigate to the CRM Dashboard
5. Confirm that the dashboard loads correctly and all modules are accessible
6. Test navigation and interaction consistency works across all devices and operating systems
 - a. Menus
 - b. Links
 - c. Buttons
7. Confirm interface renders correctly on different screen sizes
8. Confirm user permission and data access remain consistent across devices, operating systems, and connection types (on-site or remote with VPN)

Expected results:

- CRM is accessible and responsive on all approved devices and operating systems
- Logins were successful both with on-site and remote VPN connections without errors
- CRM dashboard loads correctly, and all modules are accessible on all approved devices and operating systems
- Navigational functions are consistent across all approved devices and operating systems
- Interface renders correctly on different screen sizes
- User permissions and data access are consistent across all approved devices, operating systems, and connection types

Pass/Fail:

Pass: The CRM is accessible and responsive on all approved devices, operating systems, and connection types. Logins are completed without error with both on-site and remote access via VPN. The dashboard loads correctly, and all modules are accessible. Navigational elements, user permissions, and data access remain consistent across all devices, operating systems, and connection types. The interface is consistent and renders correctly across all screen sizes.

TICKET LOGGING

Requirement to be tested:

The CRM must accurately record all actions taken on a ticket. Each entry in the ticket log must include a unique log ID linked to a valid ticket ID. Updates will be timestamped and include edits, comments, status updates, and other actions taken.

Preconditions:

- Two test users with permission to create, edit, and update tickets
- Access to the ticket logging feature of CRM

Steps:

1. Log in to the CRM as “test user 1”
2. Navigate to the “Tickets” section of the CRM dashboard
3. Create a new test ticket and enter the required details
4. Verify the following:
 - a. The ticket is assigned a unique ticket ID
 - b. The ticket log contains an entry for the ticket creation
 - c. The log entry has a unique log ID
 - d. The log entry is assigned to “test user 1”
 - e. The log entry has an accurate timestamp for when the ticket was created
5. Make the following changes and verify each change appears in the log with a unique log ID, an accurate timestamp, and is assigned to “test user 1”:
 - a. Change the status to “in progress”
 - b. Change the priority to “urgent”
 - c. Reassign the ticket to “test user 2” and save changes
6. Log out as “test user 1” and log in as “test user 2”
7. Navigate to the “Tickets” section of the CRM dashboard
8. Open the test ticket created by “test user 1”
9. Make the following changes and verify each change appears in the log with a unique log ID, an accurate timestamp, and is assigned to “test user 2”:
 - a. Update the ticket description
 - b. Change the ticket type
 - c. Change the status to “closed”
 - d. Close the ticket and save changes

Expected results:

- A unique ticket ID is generated for the new test ticket created by “test user 1”
- Each modification made to the ticket results in a corresponding entry in the ticket log with a new unique log ID
- Each entry in the ticket log has the correct user, timestamp, and type of action
- Log entry correctly shows the transfer of the ticket from “test user 1” to “test user 2”
- The final log entry for closing the ticket shows the correct information
- No duplicate or missing log entries for any modification made during the testing process



Pass/Fail:

Pass: The ticket log accurately records all ticket actions. Each modification creates a new log entry with a unique log ID, an accurate timestamp, and the user. The log shows the ticket transfer from “test user 1” to “test user 2”. The log shows the final closure of the ticket. No duplicate or missing entries were found during the testing process.

F. SOURCES**REFERENCES**

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